

CHAPTER: 3

THE Z-TRANSFORM

- used in the analysis of signals & linear time invariant systems.
- It is an extension of continuous time Fourier transform.
- can be applied to perform transfer analysis of unstable system.
- Z-transform is the discrete time analysis of the Laplace transform.

Let us consider a discrete time LTI system with impulse response $h(n)$ and input as a complex exponential signal of the form z^n . Then the o/p can be expressed as

$$y(n) = H(z) \cdot z^n \quad \rightarrow (1)$$

where,

$$H(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n} \quad \rightarrow (2)$$

If $z = e^{j\omega}$ ($\omega = \text{real number}$), then $|z| = 1$, and the summation in eqn (2) results in discrete time Fourier transform of $h(n)$.

Usually, if $|z|$ is not limited to unity, the summation is then called as the Z transform of $h(n)$.

$$X(z) = \sum_{n=-\infty}^{\infty} h(n) z^{-n}$$

$z = \text{complex variable}$

The inverse procedure, to obtain $x(n)$ from $X(z)$ is known as the "inverse z -transform".

$$X(z) = \sum \{x(n)\} \rightarrow z\text{-transform}$$

$$x(n) = Z^{-1} \{X(z)\} \rightarrow \text{Inverse } Z \text{ transform}$$

Since the Z -transform is an infinite power series, it exists only for those values of z for which this converges. The region of convergence (ROC) of $X(z)$ is the set of all the values of ' z ' for which the $X(z)$ attains a finite value.

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n}$$

$$z = re^{j\omega}$$

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) (re^{j\omega})^{-n}$$

$$= \sum_{n=-\infty}^{\infty} \{x(n) r^{-n}\} e^{-j\omega n}$$

In above equation, if $r=1$, then the z -transform evaluated on the unit circle produces the discrete time Fourier transform.

IMPORTANT POINTS

$$\rightarrow Z\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n)z^{-n}$$

$z = \text{Complex Variable}$

$\rightarrow$ If $x(n)$ is causal, i.e. $x(n) = 0$ for $n < 0$, then the z transform is called "one sided z transform"

For causal $x(n)$,

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

$\rightarrow$ If $x(n)$ is anti-causal i.e. $x(n) = 0$ for $n \geq 0$, then

$$X(z) = \sum_{n=-\infty}^{-1} x(n)z^{-n}$$

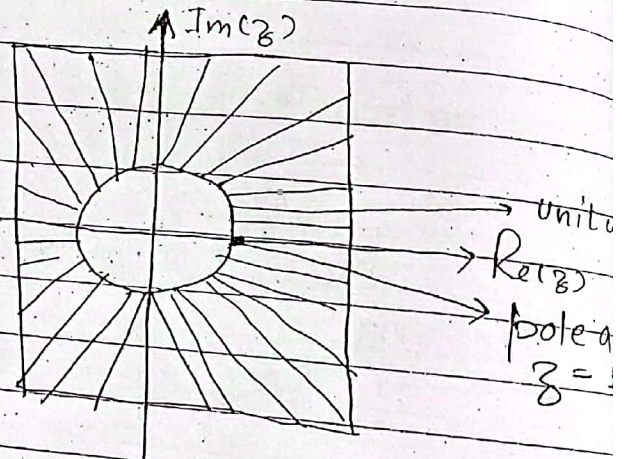
It is also known as "one sided z transform"

$\rightarrow$ The inverse z transform is used to obtain original time domain discrete signal $x(n)$ from complex domain $X(z)$.

$$x(n) = Z^{-1}[X(z)]$$

REGION OF CONVERGENCE

The set of values of 'z' in 'z-plane' for which the magnitude of $X(z)$ is finite, is called the Region of Convergence.



IMPORTANT NOTES ON "ROC"

ROC consists of a ring or circle in the z-plane centered about the origin.

ROC does not contain any pole.

$X(z)$ converges uniformly if and only if the ROC includes the unit circle, or if $\{x(n)r^{-n}\}$ is always absolutely summable

$$\sum_{n=-\infty}^{\infty} |x(n)r^{-n}| < \infty$$

If $x(n)$ is of finite duration, then ROC will be the entire z -plane except $z=0$ & $z=\infty$

If $x(n)$ is anticausal & is of finite duration, then the ROC will not include infinity (∞)

If $x(n)$ is of finite duration & causal, then the ROC will not include ' $z=0$ '

If $x(n)$ is a two sided sequence, & if the circle $|z| = r_0$ is in the ROC, then the ROC will consist of a ring in the z -plane that includes $|z| = r_0$

If the z -transform $X(z)$ is rational, then the ROC will extend to infinity

NUMERICALS.

Find the 'Z' transform of $x(n) = a^n u(n)$.

$$\begin{aligned} \therefore X(z) &= \sum_{n=-\infty}^{\infty} x(n) \cdot z^{-n} \\ &= \sum_{n=-\infty}^{\infty} a^n \cdot u(n) \cdot z^{-n} \end{aligned}$$

We know that,

$$u(n) = \begin{cases} 1, & \text{for } n \geq 0 \\ 0, & \text{otherwise.} \end{cases}$$

Hence,

$$X(z) = \sum_{n=0}^{\infty} a^n \cdot z^{-n}$$

$$= a^0 \cdot z^{-0} + a^1 \cdot z^{-1} + a^2 \cdot z^{-2} + \dots$$

$$= 1 + az^{-1} + a^2 z^{-2} + a^3 z^{-3} + \dots$$

We also know that,

$$1 + x + x^2 + x^3 + \dots = \frac{1}{1-x} \quad \text{if } |x| < 1$$

Thus,

$$X(z) = \frac{1}{1 - az^{-1}} \quad \text{if } |az^{-1}| < 1$$

$$= \frac{z}{z - a} \quad \text{if } |z| > |a|$$

Find the Z-transform of $x(n) = \delta(n)$.

We know that,

$$\delta(n) = 1, n=0 \\ = 0, \text{ otherwise.}$$

So,

$$X(z) = \sum_{n=-\infty}^{\infty} \delta(n) z^{-n}$$

because, $\delta(n)$ = causal

$$= \sum_{n=0}^{\infty} \delta(n) z^{-n}$$

$$= \delta(0) z^{-0} + \delta(1) z^{-1} + \delta(2) z^{-2} + \dots$$

$$= 1 + 0 + 0 + \dots$$

$$= 1$$

Thus,

$$X(z) = 1 \text{ for all } z$$

Find the Z-transform of discrete time unit step signal.

⇒ Here,

$$x(n) = u(n)$$

$$\therefore X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

$$\therefore u(n) = \begin{cases} 1, n \geq 0 \\ 0, \text{ otherwise} \end{cases}$$

And also, $u(n)$ = causal signal

So,

$$X(z) = \sum_{n=0}^{\infty} z^{-n}$$

$$X(z) = z^0 + z^{-1} + z^{-2} + z^{-3} + \dots$$

$$= 1 + \frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots$$

$$= \frac{1}{1 - \frac{1}{z}}, \text{ for } \left| \frac{1}{z} \right| < 1$$

$$= \frac{z}{z-1}, \text{ for } |z| > 1$$

$$\because 1 + x + x^2 + x^3 + \dots = \frac{1}{1-x}$$

if $|x| < 1$

PROPERTIES OF Z-TRANSFORM.

1. LINEARITY:

If,

$$x_1(n) \xrightarrow{Z} X_1(z)$$

$$x_2(n) \xrightarrow{Z} X_2(z)$$

then,

$$a_1 x_1(n) + a_2 x_2(n) \xrightarrow{Z} X(z) = a_1 X_1(z) + a_2 X_2(z)$$

Numericals:

a.) find Z-transform of $x(n) = \delta(n+1) + 2\delta(n) - 2\delta(n-4)$

$\Rightarrow$ from linearity property, we know that

$$X(z) = a_1 Z \{ \delta(n+1) \} + a_2 Z \{ \delta(n) \} + a_3 Z \{ \delta(n-4) \}$$

or,

$$X(z) = z \{ \delta(n+1) \} + 2 z \{ \delta(n) \} - 2 z \{ \delta(n-4) \}$$

We know that,

$$\delta(n) \xleftrightarrow{z} 1$$

$$\delta(n-k) \xleftrightarrow{z} z^{-k}$$

$$\delta(n+k) \xleftrightarrow{z} z^k$$

Using above three z -transform pairs,

$$\mathcal{Z}\{\delta(n+1)\} = z$$

$$\mathcal{Z}\{\delta(n)\} = 1$$

$$\mathcal{Z}\{\delta(n-4)\} = z^{-4}$$

Thus,

$$X(z) = z + 2 \times 1 - 4 \times z^{-4}$$

$$= z + 2 - 4z^{-4}$$

b) Find $X(z)$ if $x(n) = \cos \omega_0 n$ for $n \geq 0$.

Here,

$$x(n) = \cos \omega_0 n \text{ for } n \geq 0$$

We know that,

$$\cos \omega_0 n = \frac{1}{2} [e^{j\omega_0 n} + e^{-j\omega_0 n}]$$

So,

$$x(n) = \frac{1}{2} [e^{j\omega_0 n} + e^{-j\omega_0 n}]$$

The z -transform is expressed as,

$$X(z) = \mathcal{Z}\{x(n)\} = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Because the signal is causal, i.e. $x(n)$ is available only

for $n \geq 0$,

$$X(z) = \sum_{n=0}^{\infty} x(n) z^{-n}$$

$$= \sum_{n=0}^{\infty} \frac{1}{2} [e^{j\omega_0 n} + e^{-j\omega_0 n}] z^{-n}$$

Using linearity property, we can rewrite it as

$$X(z) = \frac{1}{2} \sum_{n=0}^{\infty} e^{j\omega_0 n} z^{-n} + \frac{1}{2} \sum_{n=0}^{\infty} e^{-j\omega_0 n} z^{-n}$$

Let,

$$x_1(n) = e^{j\omega_0 n} \quad \& \quad x_2(n) = e^{-j\omega_0 n}$$

so

$$X(z) = \frac{1}{2} \sum_{n=0}^{\infty} x_1(n) z^{-n} + \frac{1}{2} \sum_{n=0}^{\infty} x_2(n) z^{-n} \quad \rightarrow (1)$$

Let us find the 'z' transform of $x_1(n)$.

$$\begin{aligned} \mathcal{Z}\{x_1(n)\} &= \sum_{n=0}^{\infty} x_1(n) z^{-n} \\ &= \sum_{n=0}^{\infty} e^{j\omega_0 n} z^{-n} \end{aligned}$$

so,

$$\mathcal{Z}\{x_1(n)\} = \frac{1}{1 - e^{j\omega_0} z^{-1}} \quad \text{for } |z| > |e^{j\omega_0}|$$

$$[\because \mathcal{Z}\{a^n\} = \frac{1}{1 - az^{-1}}; |z| > |a|]$$

Hence,

$$\mathcal{Z}\{x_1(n)\} = \frac{1}{1 - e^{j\omega_0} z^{-1}} \quad |z| > 1$$

$$\therefore |e^{j\omega_0}| = 1$$

Similarly,

$$\mathcal{Z}\{x_2(n)\} = \sum_{n=0}^{\infty} e^{-j\omega_0 n} z^{-n}$$

$$= \frac{1}{1 - e^{-j\omega_0} z^{-1}} \quad |z| > 1$$

Using these values in eqn (1)

$$X(z) = \frac{1}{2} \cdot \frac{1}{(1 - e^{j\omega_0} z^{-1})} + \frac{1}{2} \cdot \frac{1}{(1 - e^{-j\omega_0} z^{-1})}$$

$$= \frac{1}{2} \left[\frac{1 - e^{-j\omega_0} z^{-1} + 1 - e^{j\omega_0} z^{-1}}{(1 - e^{j\omega_0} z^{-1})(1 - e^{-j\omega_0} z^{-1})} \right]$$

$$= \frac{1}{2} \left[\frac{2 - (e^{j\omega_0} + e^{-j\omega_0}) z^{-1}}{1 - e^{j\omega_0} z^{-1} - e^{-j\omega_0} z^{-1} + z^{-2}} \right]$$

$$= \frac{1 - \frac{1}{2}(e^{j\omega_0} + e^{-j\omega_0}) z^{-1}}{1 - (e^{j\omega_0} + e^{-j\omega_0}) z^{-1} + z^{-2}}$$

$$= \frac{1 - \cos \omega_0 z^{-1}}{1 - 2 \cos \omega_0 z^{-1} + z^{-2}}$$

taking z^{-2} common from denominator,

$$= \frac{1 - \cos \omega_0 z^{-1}}{z^{-2} (z^2 - 2 \cos \omega_0 z + 1)}$$

$$= \frac{z^2 - \cos \omega_0 z}{1 + z^2 - 2 \cos \omega_0 z}, \text{ for } |z| > 1.$$

2. TIME REVERSAL

If,

$$x(n) \xleftrightarrow{z} X(z) \quad \text{Roc: } r_1 < |z| < r_2$$

then,

~~$$x(n-n_0) \xleftrightarrow{z} X(z) e^{-jn_0}$$~~

$$x(-n) \xleftrightarrow{z} X(z^{-1}) \quad \text{Roc: } \frac{1}{r_2} < |z| < \frac{1}{r_1}$$

Numerical

obtain the z-transform of $x(n) = u(-n)$

$$\Rightarrow \therefore X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Let's find the 'z' transform of $u(n)$.

We know that,

$$u(n) \xleftrightarrow{z} \frac{1}{1-z^{-1}} \quad \text{Roc: } |z| > 1$$

To find the 'z' transform of $u(-n)$, we can use time reversal property:

Thus,

$$Z\{u(-n)\} \longleftrightarrow \frac{1}{1-(z^{-1})^{-1}} \quad \text{Roc: } |z| < 1$$

$$= \frac{1}{1-z} \quad \text{Roc: } |z| < 1$$

5. TIME SHIFTING PROPERTY.

If $x(n) \xleftrightarrow{z} X(z)$

then

$$x(n-n_0) \xleftrightarrow{z} z^{-n_0} X(z).$$

The region of convergence (ROC) will remain the same except for $z=0$ if $n_0 > 0$ & $z=\infty$ if $n_0 < 0$.

Proof:

We know that;

$$\sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

For $x(n-n_0)$;

$$\sum_{n=-\infty}^{\infty} x(n-n_0) z^{-n}$$

Let, $m = n - n_0$

$$\text{So, } n = m + n_0.$$

Hence,

$$\sum_{n=-\infty}^{\infty} x(n-n_0) z^{-n} = \sum_{m=-\infty}^{\infty} x(m) z^{-(m+n_0)}$$

$$= \sum_{m=-\infty}^{\infty} x(m) \cdot z^{-m} \cdot z^{-n_0}$$

The term z^{-n_0} is a constant term, hence,

$$\sum_{n=-\infty}^{\infty} x(n-n_0) z^{-n} = z^{-n_0} \sum_{m=-\infty}^{\infty} x(m) z^{-m}$$

$$= z^{-n_0} X(z).$$

Numericals:

If $X(z) = \frac{z^{-1}}{1-3z^{-1}}$ find $x(n)$.

⇒ Here,

$$\begin{aligned} X(z) &= \frac{z^{-1}}{1-3z^{-1}} \\ &= z^{-1} \cdot \frac{1}{1-3z^{-1}} \end{aligned}$$

$$\text{Let, } X_1(z) = \frac{1}{1-3z^{-1}}$$

so,

$$X(z) = z^{-1} \cdot X_1(z)$$

$$X_1(z) = \frac{1}{1-3z^{-1}}$$

$$\therefore a^n u(n) \longleftrightarrow \frac{1}{1-az^{-1}}, \quad |a| > |z|$$

so,

$$X_1(z) \longleftrightarrow \frac{1}{1-3z^{-1}} \longleftrightarrow 3^n \cdot u(n)$$

Hence,

$$X(z) = z^{-1} \cdot [3^n \cdot u(n)]$$

Using time shift property;

$$x(n) = 3^{(n-1)} \cdot u(n-1)$$

If $X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$ find $x(n)$.

Given,

$$X(z) = \frac{1 + \frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

$$= \frac{1}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

$$x(n) = Z^{-1} \left[\frac{1}{1 - \frac{1}{2}z^{-1}} \right] + Z^{-1} \left[\frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}} \right]$$

$$= Z^{-1} [x_1(z)] + Z^{-1} [x_2(z)]$$

where;

$$x_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}}$$

&

$$x_2(z) = \frac{\frac{1}{2}z^{-1}}{1 - \frac{1}{2}z^{-1}}$$

Now;

$$x_1(z) = \frac{1}{1 - \frac{1}{2}z^{-1}} = Z \left\{ \left(\frac{1}{2} \right)^n \cdot u(n) \right\}$$

Similarly,

$$x_2(z) = \frac{1}{2} z^{-1} x_1(z)$$

Using time shift property.

$$X(z) = \frac{1}{2} \cdot \left[\left(\frac{1}{2}\right)^{n-1} \cdot u(n-1) \right]$$

Hence,

$$x(n) = \left(\frac{1}{2}\right)^n u(n) + \frac{1}{2} \cdot \left(\frac{1}{2}\right)^{n-1} \cdot u(n-1).$$

$$= \left(\frac{1}{2}\right)^n u(n) + \left(\frac{1}{2}\right)^n \cdot u(n-1)$$

$$= \left(\frac{1}{2}\right)^n [u(n) + u(n-1)]$$

4. SCALING PROPERTY

If

$$x(n) \longleftrightarrow X(z)$$

$$\text{Roc: } r_1 < |z| < r_2$$

then,

$$a^n x(n) \longleftrightarrow X(a^{-1}z)$$

$$\text{Roc: } |a|r_1 < |z| < |a|r_2$$

Numericals:

Find $X(z)$ if $x(n) = 2^n u(n-2)$.

$\Rightarrow$ Let us find Z transform of $u(n)$,
which is

$$u(n) \longleftrightarrow \frac{1}{1-z^{-1}}, \quad |z| > 1$$

Now, let's find the 'z' transform of $u(n-2)$.

Using time shift property.

$$\begin{aligned} \mathcal{Z}\{u(n-2)\} &\longleftrightarrow z^{-2} \cdot \frac{1}{1-z^{-1}} \\ &= \frac{z^{-2}}{1-z^{-1}} \end{aligned}$$

Now, let's find 'z' transform of $2^n u(n-2)$.

Using scaling property;

$$\begin{aligned} \mathcal{Z}\{2^n u(n-2)\} &= \frac{(2^{-1}z)^{-2}}{1-(2^{-1}z)^{-1}} \\ &= \frac{2^2 \cdot z^{-2}}{1-2 \cdot z^{-1}} \\ &= \frac{4z^{-2}}{1-2z^{-1}} \end{aligned}$$

Using scaling property, determine the 'z' transform of $x(n) = a^n \cos \omega_0 n$.

$\Rightarrow$ soln:

From previous example we know that,

$$\begin{aligned} \mathcal{Z}[\cos \omega_0 n] &= \frac{1-z^{-1}}{1-2z^{-1}\cos \omega_0 + z^{-2}} \\ &= \frac{z(z-\cos \omega_0)}{z^2 - 2z \cdot \cos \omega_0 + 1} \end{aligned}$$

Now, using the scaling property,

$$\begin{aligned} \mathcal{Z}[a^n \cos \omega_0 n] &= \frac{1 - az^{-1} \cos \omega_0}{1 - 2az^{-1} \cos \omega_0 + a^2 z^{-2}} \\ &= \frac{z(z - a \cos \omega_0)}{z^2 - 2az \cos \omega_0 + a^2} \end{aligned}$$

5. DIFFERENTIATION PROPERTY.

If,

$$x(n) \xleftrightarrow{z} X(z)$$

then,

$$n x(n) \xleftrightarrow{z} -z \frac{dX(z)}{dz}$$

or,

$$n x(n) \xleftrightarrow{z} z^{-1} \frac{dX(z)}{dz^{-1}}$$

Numericals:

a.) Determine $X(z)$ if $x(n) = n^2 u(n)$.

Soln:

$$x(n) = n^2 u(n)$$

$$= n \cdot [n u(n)]$$

Using differentiation property,

$$X(z) = z^{-1} \frac{dX(z)}{dz^{-1}}$$

$$X(z) = z^{-1} \frac{d}{dz^{-1}} \{ z^{-1} \sum_{n=0}^{\infty} u(n) z^{-n} \}$$

Again using differentiating property for the factor in the bracket, we get

$$X(z) = z^{-1} \cdot \frac{d}{dz^{-1}} \cdot z^{-1} \frac{d}{dz^{-1}} \frac{1}{1-z^{-1}}$$

$$\text{or, } X(z) = z^{-1} \frac{d}{dz^{-1}} \cdot z^{-1} \cdot \frac{1}{(1-z^{-1})^2}$$

$$\left[\because d\left(\frac{1}{v}\right) = \frac{-v'}{v^2} \right]$$

$$= z^{-1} \cdot \frac{d}{dz^{-1}} \cdot \frac{z^{-1}}{(1-z^{-1})^2}$$

$$\left[\because d\left(\frac{u}{v}\right) = \frac{u'v - v'u}{v^2} \right]$$

$$= z^{-1} \cdot \left[\frac{(1-z^{-1})^2 - z^{-1} \cdot 2(1-z^{-1})(-1)}{(1-z^{-1})^4} \right]$$

$$= z^{-1} \cdot \frac{(1-z^{-1})(1-z^{-1}+2z^{-1})}{(1-z^{-1})^4}$$

$$= z^{-1} \cdot \frac{(1+z^{-1})}{(1-z^{-1})^3}$$

Find $x(n)$ if $X(z) = \log(1+az^{-1})$, $|z| > |a|$

Given;

$$X(z) = \log(1+az^{-1})$$

Taking differentiation;

$$\frac{dX(z)}{dz} = \frac{-az^{-2}}{1+az^{-1}}$$

Multiplying both sides by $-z$;

$$z \cdot \frac{dX(z)}{dz} = -z \cdot \left(\frac{-az^{-2}}{1+az^{-1}} \right)$$

$$= \frac{a \cdot z^{-1}}{1+az^{-1}} \quad |z| > |a|$$

$$= a \cdot z^{-1} \cdot \left[\frac{1}{1-(-a)z^{-1}} \right]$$

Using time delay property

$$z \cdot \frac{dX(z)}{dz} = a \cdot (-a)^{n-1} u(n-1)$$

But we know that;

$$z \cdot \frac{dX(z)}{dz} \rightarrow n \cdot x(n)$$

(21)

So,

$$nx(n) = a \cdot a^{n-1} \cdot u(n-1)$$

$$\text{or, } nx(n) = a \cdot u(n-1)$$

$$x(n) = \frac{a \cdot 1 \cdot u(n-1)}{n}$$

6. CONVOLUTION PROPERTY

If,

$$x_1(n) \xleftrightarrow{Z} X_1(z)$$

$$\& x_2(n) \xleftrightarrow{Z} X_2(z)$$

Then,

$$x_1(n) \otimes x_2(n) \xleftrightarrow{Z} X_1(z) \cdot X_2(z)$$

Numericals:

Compute the convolution $x(n)$ of the signals

$$x_1(n) = \{4, -2, 1\}$$

$$x_2(n) = \begin{cases} 1, & \text{for } 0 \leq n \leq 3 \\ 0, & \text{otherwise} \end{cases}$$

$\Rightarrow$ The 'z' transforms for the given signals can be written as.

$$X_1(z) = 4 - 2z^{-1} + z^{-2}$$

$$X_2(z) = 1 + z^{-1} + z^{-2} + z^{-3}$$

Therefore,

$$X(z) = X_1(z) \cdot X_2(z)$$

$$= (4 - 2z^{-1} + z^{-2})(1 + z^{-1} + z^{-2} + z^{-3})$$

$$= 4 + 4z^{-1} + 4z^{-2} + 4z^{-3} - 2z^{-1} - 2z^{-2} - 2z^{-3} - 2z^{-4}$$

$$= 4 + z^{-2} + z^{-3} + z^{-4} + z^{-5}$$

$$X(z) = 4 + 2z^{-1} + 3z^{-2} + 3z^{-3} - z^{-4} + z^{-5}$$

So,

$$x(n) = \{4, 2, 3, 3, -1, 1\}$$

Determine $x(n)$ by using convolution for

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})}$$

Given;

$$X(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})(1 + \frac{1}{4}z^{-1})}$$

$$= X_1(z) \cdot X_2(z)$$

where,

$$X_1(z) = \frac{1}{(1 - \frac{1}{2}z^{-1})}$$

$$\& X_2(z) = \frac{1}{(1 + \frac{1}{4}z^{-1})}$$

taking 'z' inverse, we get

$$x_1(n) = \left(\frac{1}{2}\right)^n \cdot u(n)$$

&

$$x_2(n) = \left(-\frac{1}{4}\right)^n \cdot u(n)$$

We know,

$$x(n) = x_1(n) \otimes x_2(n)$$

Since the signals are causal,

$$x_1(n) = 1, \quad n=0$$

$$x(n) = \sum_{k=0}^n x_1(n-k) \cdot x_2(k)$$

$$= \sum_{k=0}^n \left(\frac{1}{2}\right)^{n-k} \cdot \left(\frac{-1}{4}\right)^k$$

$$= \left(\frac{1}{2}\right)^n \sum_{k=0}^n \left(\frac{-1/4}{1/2}\right)^k$$

$$= \left(\frac{1}{2}\right)^n \cdot \sum_{k=0}^n \left(\frac{-1}{2}\right)^k$$

$$= \left(\frac{1}{2}\right)^n \cdot \frac{1 - (-1/2)^{n+1}}{1 - (-1/2)}$$

$$= \left(\frac{1}{2}\right)^n \cdot \frac{2}{3} \left[1 - \left(\frac{-1}{2}\right) \left(\frac{-1}{2}\right)^n \right]$$

$$= \left[\frac{2}{3} \left(\frac{1}{2}\right)^n + \frac{1}{3} \left(\frac{-1}{4}\right)^n \right] u(n)$$

7. CORRELATION PROPERTY

If,

$$x_1(n) \xleftrightarrow{z} X_1(z)$$

$$x_2(n) \xleftrightarrow{z} X_2(z)$$

then,

$$r_{x_1 x_2}(l) = \sum_{n=-\infty}^{\infty} x_1(n) \cdot x_2(n-l)$$

$$r_{x_1 x_2}(l) \xleftrightarrow{z} X_1(z) \cdot X_2(z^{-1})$$

Determine the cross-correlation of

$$x_1(n) = \{1, 2, 3, 4\}$$

$$x_2(n) = \{4, 3, 2, 1\}$$

soln:

$$X_1(z) = 1 + 2z^{-1} + 3z^{-2} + 4z^{-3}$$

$$X_2(z) = 4 + 3z^{-1} + 2z^{-2} + z^{-3}$$

so,

$$X_2(z^{-1}) = 4 + 3z + 2z^2 + z^3$$

Now,

$$R_{x_1 x_2}(z) = X_1(z) \cdot X_2(z^{-1})$$

$$= (1 + 2z^{-1} + 3z^{-2} + 4z^{-3})(4 + 3z + 2z^2 + z^3)$$

$$= z^3 + 4z^2 + 10z + 20 + 25z^{-1} + 24z^{-2} + 16z^{-3}$$

Hence,

$$r_{x_1 x_2}(l) = \{1, 4, 10, 20, 25, 24, 16\}$$

8. CONJUGATION PROPERTY

If,

$$x(n) \xleftrightarrow{z} X(z)$$

Then,

$$x^*(n) \xleftrightarrow{z} X^*(z^*)$$

If, $x(n)$ is real then,

$$X(z) = X^*(z^*)$$

This means that if z -transform $X(z)$ has a pole or zero at $z = z_0$, then it must have a pole or zero at the complex conjugate point $z = z_0^*$.

9. INITIAL VALUE THEOREM.

It states that if $x(n)$ is a causal discrete time signal with z transform $X(z)$, then the initial value may be determined by using the following expression

$$x(0) = \lim_{n \rightarrow 0} x(n) = \lim_{|z| \rightarrow \infty} X(z).$$

10. FINAL VALUE THEOREM.

The final value theorem states that for a discrete time signal $x(n]$, if $X(z)$ and the poles of $X(z)$ are all inside the unit circle, then the final value of the discrete time signal, $x(\infty)$, may be determined by using the following expression

$$x(\infty) = \lim_{n \rightarrow \infty} x(n) = \lim_{|z| \rightarrow 1} [(1-z^{-1})X(z)]$$

Numericals

If $X(z) = 2 + 3z^{-1} + 4z^{-2}$. Find initial value & the final value of $x(n)$.

Solⁿ:

$$X(z) = 2 + 3z^{-1} + 4z^{-2}$$

We know that initial value is given by,

$$x(0) = \lim_{n \rightarrow 0} x(n) = \lim_{|z| \rightarrow \infty} X(z)$$

$$= \lim_{|z| \rightarrow \infty} (2 + 3z^{-1} + 4z^{-2})$$

$$= \lim_{|z| \rightarrow \infty} \left(2 + \frac{3}{z} + \frac{4}{z^2} \right)$$

$$= \frac{2}{\infty} + \frac{3}{\infty} + \frac{4}{\infty}$$

$$= 2 + 0 + 0$$

$$= 2$$

And, final value is

$$x(\infty) = \lim_{n \rightarrow \infty} x(n) = \lim_{|z| \rightarrow 1} [(1 - z^{-1}) X(z)]$$

$$\text{So, } x(\infty) = \lim_{|z| \rightarrow 1} [(1 - z^{-1}) \cdot (2 + 3z^{-1} + 4z^{-2})]$$

$$= \lim_{|z| \rightarrow 1} [2 + z^{-1} + z^{-2} - 4z^{-3}]$$

$$= 2 + 1 + 1 - 4$$

$$= 0$$

11. TIME DELAY (FOR ONE SIDED Z-TRANSFORM)

If,

$$x(n) \xleftrightarrow{z} X(z), \text{ then}$$

$$x(n-k) \xleftrightarrow{z} z^{-k} \left[X(z) + \sum_{n=1}^k x(-n) z^n \right] \quad k > 0$$

12. TIME ADVANCE PROPERTY

If,

$$x(n) \xleftrightarrow{z} X(z)$$

then,

$$x(n+k) \xleftrightarrow{z} z^k \left[X(z) - \sum_{n=0}^{k-1} x(n) z^{-n} \right]$$

THE INVERSE Z-TRANSFORM

The inverse z-transform is expressed as

$$x(n) = Z^{-1}[X(z)]$$

There are three methods to obtain inverse z-transform

- i) Long Division Method
- ii) Partial Fraction Method
- iii) Residue Method

1. LONG DIVISION METHOD

$$X(z) = \sum_{n=0}^{\infty} x(n)z^{-n}$$

We know that,

$$X(z) = \frac{N(z)}{D(z)}$$

The above ratio of polynomials for z-transform may be divided out to produce a power series in the form of an equation with the coefficients representing the sequence values in the time domain as

$$X(z) = \frac{N(z)}{D(z)} = \sum_{n=0}^{\infty} a_n z^{-n}$$

Here,

$$= a_0 z^0 + a_1 z^{-1} + a_2 z^{-2} + \dots$$

the coefficients a_n are the values of $x(n)$.

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The region of convergence determines whether the series has positive or negative exponents. Causal sequence will have exponents negative whereas anti causal sequence will have positive exponents.

NUMERICALS:

If $h(n) = \{1, 2, 3\}$ & $y(n) = \{1, 1, 2, -1, 3\}$. Find $x(n)$.

Soln:

$$H(z) = \mathcal{Z}\{h(n)\} = 1 + 2z^{-1} + 3z^{-2}$$

$$Y(z) = \mathcal{Z}\{y(n)\} = 1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}$$

We know;

$$\begin{aligned} X(z) &= \frac{Y(z)}{H(z)} \\ &= \frac{1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}}{1 + 2z^{-1} + 3z^{-2}} \end{aligned}$$

Now,

$$\begin{array}{r} 1 - z^{-1} + z^{-2} \\ 1 + 2z^{-1} + 3z^{-2} \overline{) 1 + z^{-1} + 2z^{-2} - z^{-3} + 3z^{-4}} \\ \underline{1 + 2z^{-1} + 3z^{-2}} \phantom{- z^{-3} + 3z^{-4}} \\ -z^{-1} - z^{-2} - z^{-3} + 3z^{-4} \\ \underline{-z^{-1} - 2z^{-2} - 3z^{-3}} \phantom{+ 3z^{-4}} \\ z^{-2} + 2z^{-3} + 3z^{-4} \\ \underline{z^{-2} + 2z^{-3} + 3z^{-4}} \\ 0 \end{array}$$

X.

Thus,

$$X(z) = 1 - z^{-1} + z^{-2}.$$

The ~~inverse~~ inverse z -transform of above expression,
 $x(n) = \{1, -1, 1\}$.

Using long division, determine the inverse z -transform
of $X(z) = \frac{1}{1 - (3/2)z^{-1} + (1/2)z^{-2}}$.

a.) Roc: $|z| > 1$

b.) Roc: $|z| < 1/2$

Soln:

a.) When $|z|$ Roc: is greater than '1', it indicates that the Roc is the exterior of a circle, therefore $x(n)$ is a causal signal. Hence, we need a power series expansion in negative power of z . By dividing the numerator of $X(z)$ by its denominator, we get

(Continued in next page).

$$\begin{array}{r}
 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \dots \\
 1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2} \overline{) 1} \\
 \underline{1 - \frac{3}{2}z^{-1} + \frac{1}{2}z^{-2}} \\
 \frac{3}{2}z^{-1} - \frac{1}{2}z^{-2} \\
 \underline{\frac{3}{2}z^{-1} - \frac{9}{4}z^{-2} + \frac{3}{4}z^{-3}} \\
 \frac{7}{4}z^{-2} - \frac{3}{4}z^{-3} \\
 \underline{\frac{7}{4}z^{-2} - \frac{21}{8}z^{-3} + \frac{7}{8}z^{-4}} \\
 \frac{15}{8}z^{-3} - \frac{7}{8}z^{-4}
 \end{array}$$

Therefore

$$X(z) = 1 + \frac{3}{2}z^{-1} + \frac{7}{4}z^{-2} + \dots$$

Hence,

$$x(n) = \left\{ 1, \frac{3}{2}, \frac{7}{4}, \dots \right\}$$

b.) If $Roc: |z| < \frac{1}{2}$, it indicates that the Roc is the interior of a circle. So the sequence is anti-causal. To obtain a power series expansion in positive powers of z , so we shall perform the long division in the following way,

$$\begin{array}{r}
 2z^2 + 6z^3 + 14z^4 + 30z^5 + 62z^6 + \dots \\
 \underline{1z^{-2} - \frac{3}{2}z^{-1} + 1)} \quad 1 \\
 1 - 3z + 2z^2 \\
 3z - 2z^2 \\
 \underline{3z - 9z^2 + 6z^3} \\
 7z^2 - 6z^3 \\
 \underline{7z^2 - 21z^3 + 14z^4} \\
 15z^3 - 14z^4 \\
 \underline{15z^3 - 45z^4 + 30z^5} \\
 31z^4 - 30z^5
 \end{array}$$

Therefore,

$$X(z) = 2z^2 + 6z^3 + 14z^4 + 30z^5 + 62z^6 + \dots$$

Hence,

$$\begin{array}{l}
 \text{and from eqn (4),} \\
 x(n) = \{ \dots, 6, 2, 8, 0, 14, 6, 2, 0, 0 \}
 \end{array}$$

2. PARTIAL FRACTION EXPANSION METHOD

With the help of partial fraction expansion method, we convert a system transfer function into a sum of standard functions. To do this, firstly, the denominator of the transfer function is $H(z)$ is factorised into prime factors & then simple poles of $H(z)$ is obtained.

Examples:

Use partial fraction method to find inverse z -transform of $H(z) = \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}}$

solⁿ:

$$H(z) = \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}}$$

$$= \frac{-4 + 8z^{-1}}{1 + 6z^{-1} + 8z^{-2}}$$

$$= \frac{-4 + 8z^{-1}}{(1 + 4z^{-1})(1 + 2z^{-1})}$$

which can be expressed as,

$$H(z) = \frac{A}{1 + 4z^{-1}} + \frac{B}{1 + 2z^{-1}}$$

Now,

$$A = \frac{-4 + 8z^{-1}}{1 + 2z^{-1}} \quad \Bigg| \quad \text{at } z^{-1} = -\frac{1}{4}$$

$$= \frac{-4 + 8(-1/4)}{1 + 2(-1/4)}$$

$$= \frac{-4 - 2}{1 - 1/2} = \frac{-6}{1/2}$$

$$= -12$$

$$B = \frac{-4 + 8z^{-1}}{1 + 4z^{-1}} \quad \Bigg| \quad \text{at } z^{-1} = -1/2$$

$$= \frac{-8}{-1}$$

$$= 8$$

Hence,

$$H(z) = \frac{-12}{1 + 2z^{-1}} + \frac{8}{1 + 4z^{-1}}$$

so,

$$h(n) = \left\{ -12 \left(-\frac{1}{2} \right)^n \cdot u(n) + 8 \left(-\frac{1}{4} \right)^n \cdot u(n) \right\}$$

Find the causal signal $x(n)$ which is having z -transform as

$$X(z) = \frac{z^3}{(z+1)(z-1)^2}$$

Solution;

$$X(z) = \frac{z^3}{(z+1)(z-1)^2}$$

Since, the degree of polynomial in numerator is equal to the degree of polynomial in denominator. Let's divide both sides by 'z' we get

$$\frac{X(z)}{z} = \frac{z^2}{(z+1)(z-1)^2}$$

$$\frac{X(z)}{z} = \frac{z^2}{(z+1)(z-1)^2} \rightarrow (1)$$

$$\frac{X(z)}{z} = \frac{A}{z+1} + \frac{B}{z-1} + \frac{C}{(z-1)^2}$$

$$A(z-1)^2 + B(z+1)(z-1) + C(z+1) \rightarrow (2)$$

Comparing the numerator of eqn (1) & (2) we get,

$$z^2 = A(z-1)^2 + B(z+1)(z-1) + C(z+1)$$

Let's assume $z = -1$, we get

$$(-1)^2 = A(-1-1)^2 + B(-1+1)(-1-1) + C(-1+1)$$

$$1 = 4A + 0 + 0$$

$$\therefore A = \frac{1}{4}$$

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Similarly, if $z = 1$, we get.

$$(1)^2 = A(0-1)^2 + B(1+1)(1-1) + C(1+1)$$

$$1 = 0 + 0 + 2C$$

$$\therefore C = \frac{1}{2}$$

Again if $z = 0$ we get.

$$0 = A(0-1)^2 + B(0+1)(0-1) + C(0+1)$$

$$0 = A - B + C$$

$$0 = \frac{1}{4} - B + \frac{1}{2}$$

$$B = \frac{1+2}{4} = \frac{3}{4}$$

So, finally we get

$$\cancel{X(z)} = \cancel{\frac{1/4}{(z-1)^2}} + \cancel{\frac{3/4}{z}}$$

$$X(z) = \frac{1/4}{z+1} + \frac{3/4}{z-1} + \frac{1/2}{(z-1)^2}$$

So,

$$X(z) = \frac{1}{4} \frac{z}{z+1} + \frac{3}{4} \frac{z}{z-1} + \frac{1}{2} \frac{z}{(z-1)^2}$$

$$= \frac{1}{4} (-1)^n u(n) + \frac{3}{4} (1)^n u(n) + \frac{1}{2} n u(n) \quad \left| \begin{array}{l} \frac{z}{z-1} \leftrightarrow a^n u(n) \\ \frac{z}{(z-1)^2} \leftrightarrow n a^n u(n) \end{array} \right.$$

$$= \left[\frac{1}{4} (-1)^n + \frac{3}{4} + \frac{1}{2} n \right] u(n)$$

3. RESIDUE METHOD

In this method, we obtain, inverse z -transform $x[n]$, by summing residues of $[X(z)z^{n-1}]$ at all poles. Mathematically,

$$x[n] = \sum_{\substack{\text{all poles} \\ X(z)}} \text{Residue of } [X(z)z^{n-1}]$$

Here, the residue for any poles of order m at $z = \beta$ is

$$\text{Residue} = \frac{1}{(m-1)!} \lim_{z \rightarrow \beta} \left\{ \frac{d^{m-1}}{dz^{m-1}} [(z-\beta)^m X(z) \cdot z^{n-1}] \right\}$$

CAUSALITY OF DISCRETE TIME LTI SYSTEM.

A discrete time LTI system which has a rational transfer function $H(z)$ will be causal if and only if

→ the Roc is the exterior of the circle outside the outermost pole, &

→ with $H(z)$ expressed as a ratio of polynomial in z , the order of numerator should be smaller than order of denominator.

STABILITY CRITERIA

A discrete time LTI system is stable if & only if the Roc of its transfer function $H(z)$ includes the unit circle $|z|=1$.

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